

Kamal Joshi

Staff Software Engineer, Cloud Identity | Go | OAuth 2.0 | OIDC | OPA | Authorization Systems | AWS

Austin, TX | Remote | Open to relocation for the right opportunity

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PROFESSIONAL SUMMARY

Staff software engineer specializing in production identity and authorization systems: OAuth 2.0/OIDC flows, API key management, JWT/JWKS lifecycle, OPA policy enforcement, and fine-grained authorization services — all in Go on AWS at scale. Spent a decade building and owning the identity and access infrastructure that Cisco Secure Access enterprise customers depend on, including authentication services, authorization libraries, and API gateway enforcement layers. Identified and reported a base64 URL decoding bug in a widely-used Go JWT/JWKS library ([MicahParks/compatibility-keyfunc#2](https://github.com/MicahParks/compatibility-keyfunc#2)), demonstrating genuine depth in identity protocol implementation. Fluent in AI-assisted engineering: I use GitHub Copilot and Claude Code daily to produce safe, testable, readable code while owning design, quality, and business outcomes.

CORE STRENGTHS

- **Identity and authentication:** OAuth 2.0/OIDC, JWT/JOSE, JWKS rotation, API key issuance and refresh, token lifecycle management, Secrets Manager integration, TLS certificate management
- **Authorization at scale:** OPA (Open Policy Agent) policy enforcement, RBAC, authorization services, API gateway policy, auth-tools internal library adopted across platform services
- **Production reliability:** latency-sensitive distributed systems, on-call incident ownership, DLQ recovery, multi-region HA, rate-limit governance, deployment validation
- **Cloud-native platform:** Go, AWS (ECS, EKS, S3, SQS/SNS, DynamoDB, Secrets Manager), Kubernetes, Docker, Terraform, CI/CD
- **Security and compliance:** OWASP secure coding (100+ warnings to zero), ISMAP security certification, sovereign/GovCloud environments, enterprise integrations (Microsoft Intune, Jamf Pro)
- **Technical leadership:** architecture design, cross-team IAM coordination, mentoring early-in-career engineers, customer-facing incident response, open source contributions

EXPERIENCE

Cisco

Software Engineering Technical Leader

November 2025 - Present | Austin, TX

- Owned **Authorization Service** and **API Gateway** — the identity enforcement layer for Cisco Secure Access — designing and delivering **OAuth 2.0/OIDC** token flows, **API key management**, and policy enforcement across a multi-tenant SaaS platform.
- Mapped **AuthSvc TLS secret synchronization** and **API key refresh behavior** across deployment tooling, Secrets Manager, Kubernetes, and Kong — clarifying the token lifecycle and eliminating ambiguity in security-sensitive automation paths.
- Resolved a customer-scale onboarding failure by tracing **250+ appliance registration requests** through API key auth and rate-limit behavior, driving durable fixes around retry/backoff, **token reuse**, and service ownership.
- Drove cross-service design for **Splunk** and **CloudEvents** webhook event formats across alerts, security events, integrations, and notifications services, establishing consistent auth and delivery patterns across four teams.
- **Diagnosed and fixed a critical integration issue live during a prospective customer evaluation**, contributing directly to the **customer's purchase decision**.
- **Prevented security-event message loss** during a production incident by identifying a **300k+ message DLQ backlog**, quantifying impact across **15 orgs**, and redriving messages within **5-10 minutes** — full on-call incident ownership from detection to resolution.

- Drove integrations-svc production readiness across **sovereign and multi-region environments** using Terraform, validating **HA behavior across two regions**.

Senior Software Engineer

January 2021 - November 2025 | Austin, TX

- Built **auth-tools**, an internal **authorization library** adopted across multiple platform services to simplify **OAuth/OIDC policy enforcement** integration — the kind of "secure by default" pattern that scales.
- Led **ISMAP security certification** for the APIX platform, coordinating cross-functional IAM and compliance requirements end-to-end.
- Integrated **AI tooling (GitHub Copilot, Claude Code, Codex)** into daily engineering workflow; pioneered team-wide Copilot adoption ahead of broader Cisco AI initiatives.
- Mentored early-in-career engineers through code review, design feedback, and hands-on guidance.

Randstad Technologies US at Cisco

Senior Software Developer

March 2015 - January 2021 | Dublin, OH

- Built **OIDC/OAuth 2.0** authentication and authorization flows integrated across frontend and backend services; deployed **OPA as a sidecar** for authorization policy enforcement (2020-2021).
- Designed and implemented **Go-based gRPC and JSON/HTTP APIs** for a cloud-native data ingestion platform.
- Filed [MicahParks/compatibility-keyfunc#2](#): identified a **base64 URL decoding bug in RSA key modulus parsing** in a widely-used Go JWT/JWKS library — acknowledged and fixed by the maintainer.
- Reduced **100+ OWASP-reported security warnings to zero** through secure coding, browser security headers, input sanitization, and server-side validation.

Earlier Engineering Experience

Cognizant, Oracle, Collabera | January 2007 - February 2015

- Delivered backend services and integrations across financial services, healthcare, insurance, and enterprise domains.
- Built foundation in Java, REST APIs, service integrations, and maintainable engineering practices.

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate
- AWS Certified Developer - Associate
- MongoDB Certified Java Developer M101J
- Google Cloud Platform Essentials

EDUCATION

Bachelor of Engineering, Information Technology
Dharmsinh Desai University